King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 201 Final Exam Term 223

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. The minimum value of the function $f(x, y) = 3x^2 - y^2$ subject to the constraint $2x - 2y + 5 = 0$ is equal to

(a) $-\dfrac{75}{8}$

(b) $-\dfrac{1}{4}$

(c) $-\dfrac{225}{4}$

(d) $-\dfrac{50}{3}$

(e) $-\dfrac{25}{4}$

2. $\int_1^4 \int_0^1 \int_0^x 2z\, e^{-x^2}\, dy\, dx\, dz =$

(a) $\dfrac{9(e^{-1} - 1)}{2}$

(b) $\dfrac{7(1 - e^{-1})}{3}$

(c) $\dfrac{15(1 - e^{-1})}{2}$

(d) $\dfrac{e^{-1} - 1}{3}$

(e) $8(1 - e^{-1})$

3. If the slope of the tangent line to the parametric curve $x = 3t^2 + 1$, $y = t^3 - 1$ at the point (a, b) is equal to $\dfrac{1}{2}$, then $a + b =$

(a) 4

(b) $\dfrac{1}{2}$

(c) -1

(d) 1

(e) 0

4. An equation in rectangular coordinates for the surface represented by the cylindrical equation $r^2 \cos 2\theta - z^2 + 1 = 0$ is

(a) $-x^2 + y^2 + z^2 = 1$

(b) $x^2 + y^2 - z^2 = 1$

(c) $x^2 - y^2 - z^2 = 1$

(d) $x^2 + y^2 + z^2 = 1$

(e) $x^2 - y^2 + z^2 = 1$

5. The projection of the vector $\vec{u} = \langle 0, 3, 3 \rangle$ onto the vector $\vec{v} = \langle -1, 1, 1 \rangle$ is

(a) $\langle 2, 2, 2 \rangle$

(b) $\langle -2, 2, -2 \rangle$

(c) $\langle -2, -2, 2 \rangle$

(d) $\langle -2, 2, 2 \rangle$

(e) $\langle 2, 2, -2 \rangle$

6. The volume of the solid in the first octant bounded by the coordinate planes and the plane $z = 7 - x - 2y$ is given by

(a) $\int_0^{\frac{7}{2}} \int_0^{7-2y} \int_0^{7-x-2y} dz\, dy\, dx$

(b) $\int_0^{7} \int_0^{\frac{7-x}{2}} \int_0^{7-x-2y} dz\, dy\, dx$

(c) $\int_0^{7} \int_0^{7-x} \int_0^{7-x-2y} dz\, dy\, dx$

(d) $\int_0^{7} \int_0^{\frac{7}{2}} \int_0^{7-x-2y} dz\, dy\, dx$

(e) $\int_0^{7} \int_0^{\frac{7}{2}} \int_0^{7} dz\, dy\, dx$

7. If $f(x, y, z) = z \sin(xy^2 + 2z)$, then $\dfrac{\partial f}{\partial z}(1, -1, -1) =$

(a) $-2\cos(1) - \sin(1)$

(b) $2\cos(1) - \sin(1)$

(c) $-2\cos(1) + \sin(1)$

(d) $2\cos(1) - 2\sin(1)$

(e) $2\cos(1) + \sin(1)$

8. The tangent line to the curve of intersection of the surfaces $z = \sqrt{x^2 + y^2}$ and $x - 2y + 2z = 20$ at the point $(4, -3, 5)$ contains the point

(a) $(12, -16, 0)$

(b) $(-12, -16, 0)$

(c) $(16, 12, 0)$

(d) $(-12, 16, 0)$

(e) $(12, 16, 0)$

9. Let $I = \int_0^2 \int_0^x \sqrt{x^2 + y^2}\, dy\, dx + \int_2^{2\sqrt{2}} \int_0^{\sqrt{8-x^2}} \sqrt{x^2 + y^2}\, dy\, dx$. After writing I as a single iterated integral by converting to polar coordinates, $I =$

(a) $\dfrac{2\sqrt{2}\,\pi}{3}$

(b) $\dfrac{16\sqrt{2}\,\pi}{3}$

(c) $\dfrac{8\sqrt{2}\,\pi}{3}$

(d) $\dfrac{4\sqrt{2}\,\pi}{3}$

(e) $\dfrac{\sqrt{2}\,\pi}{3}$

10. Using cylindrical coordinates, we find that

$$\int_0^4 \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} \int_{x^2+y^2}^{16} y\, dz\, dy\, dx =$$

(a) 4π

(b) 16π

(c) 2π

(d) 0

(e) 8π

11. If $z = e^{xy}$, $x = 3s + t$ and $y = 3s - t$, then the value of $\dfrac{\partial z}{\partial s}$ when $s = \dfrac{1}{3}$ and $t = 1$ is equal to

(a) 18

(b) 12

(c) 6

(d) 9

(e) 3

12. The maximum value of the function $f(x, y) = 2x^2 + 3y^2 - 4x - 7$ on the disc $x^2 + y^2 \leq 16$ is equal to

(a) 39

(b) 43

(c) 41

(d) 45

(e) 37

13. The normal line at the point (e, e, e) to the surface $x \ln(y^2 z) = 3e$ contains the point

(a) $(e - 3,\, e + 2,\, e - 1)$

(b) $(e - 3,\, e - 2,\, e - 1)$

(c) $(e + 3,\, e + 2,\, e - 1)$

(d) $(e + 3,\, e - 2,\, e + 1)$

(e) $(e - 3,\, e - 2,\, e + 1)$

14. The rectangular equation $x^2 + y^2 + z^2 - 3z = 0$ in spherical coordinates is

(a) $\rho = 3 \cos \phi$

(b) $\rho = \sqrt{3}$

(c) $\rho = \sqrt{3} \cos \phi$

(d) $\rho = 3 \sin \phi$

(e) $\rho = \sqrt{3} \sin \phi$

15. Suppose that the equation of the plane that passes through the point $(2, 2, 1)$ and contains the line $\dfrac{x}{2} = \dfrac{y - 4}{-1} = z$ is given by $ax + by - 2z = 0$. Then $a + b =$

(a) 0

(b) 1

(c) -3

(d) -6

(e) 3

16. The volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the xy-plane and below the cone $z = \sqrt{x^2 + y^2}$ is equal to

(a) $\dfrac{\sqrt{2}\,\pi}{3}$

(b) $\dfrac{8\sqrt{2}\,\pi}{3}$

(c) $\dfrac{\sqrt{3}\,\pi}{3}$

(d) $\dfrac{5\sqrt{3}\,\pi}{2}$

(e) $\dfrac{8\pi}{3}$

17. The area of the region enclosed by the polar curve $r = 4\cos\theta$ is equal to

(a) 4π

(b) π

(c) 8π

(d) 6π

(e) 2π

18. The minimum distance from the point $(6, 6, 0)$ to the surface $z = \sqrt{10 + 6x + 6y}$ is

(a) 6

(b) 10

(c) 8

(d) 12

(e) 16

19. The function $f(x, y) = 4xy - 4x - 2y^2 + 4y - x^3 - 1$ has

(a) two relative extrema

(b) a relative minimum and a saddle point

(c) no relative extrema and no saddle points

(d) a relative maximum and a saddle point

(e) two saddle points

20. The volume of the solid bounded by the graphs of $z = \sqrt{x^2 + y^2}$, $z = 0$ and $x^2 + y^2 = 25$ is equal to

(a) $\dfrac{200\pi}{3}$

(b) $\dfrac{125\pi}{3}$

(c) $\dfrac{250\pi}{3}$

(d) $\dfrac{50\pi}{3}$

(e) $\dfrac{75\pi}{3}$

21. $\int_0^2 \int_{y^2}^4 \sqrt{x}\, \sin x \, dx \, dy =$

(a) $2\cos(2) + \sin(2)$

(b) $-4\cos(4) + \sin(4)$

(c) $4\cos(4) - \sin(4)$

(d) $2\cos(2) - \sin(2)$

(e) $\cos(4) - 4\sin(4)$

22. In spherical coordinates, the integral

$$\int_{-1}^{1} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \int_{1-\sqrt{1-x^2-y^2}}^{1+\sqrt{1-x^2-y^2}} (x^2 + y^2 + z^2)^{3/2} \, dz \, dy \, dx$$

is equal to

(a) $\int_0^{2\pi} \int_0^{\pi} \int_0^{2\cos\phi} \rho^5 \sin\phi \, d\rho \, d\phi \, d\theta$

(b) $\int_0^{2\pi} \int_0^{\pi/2} \int_0^{2\cos\phi} \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$

(c) $\int_0^{2\pi} \int_0^{\pi/2} \int_0^{2\cos\phi} \rho^5 \sin\phi \, d\rho \, d\phi \, d\theta$

(d) $\int_0^{2\pi} \int_0^{\pi/2} \int_0^{\cos\phi} \rho^5 \sin\phi \, d\rho \, d\phi \, d\theta$

(e) $\int_0^{2\pi} \int_0^{\pi} \int_0^{2\cos\phi} \rho^4 \sin\phi \, d\rho \, d\phi \, d\theta$

23. The average value of $f(x, y) = e^{x+y}$ over R, where R is the triangle with vertices $(0, 0)$, $(0, 1)$ and $(1, 1)$, is

(a) $e - 1$

(b) $e^2 - 2$

(c) $(e - 1)^2$

(d) $e + 1$

(e) $(e + 1)^2$

24. If the absolute maximum and the absolute minimum of the function $f(x, y) = x^2 - 2xy + 2y$ over the region $R = \{(x, y) \mid 0 \leq x \leq 3,\ 0 \leq y \leq 2\}$ are M and m respectively, then $M + m =$

(a) 15

(b) 16

(c) 9

(d) 12

(e) 8

25. $\int_1^{10} \int_0^{\ln y} f(x, y)\, dx\, dy =$

(a) $\int_0^{\ln 10} \int_1^{10} f(x, y)\, dy\, dx$

(b) $\int_0^{10} \int_0^{\ln x} f(x, y)\, dy\, dx$

(c) $\int_0^{\ln 10} \int_{e^x}^{10} f(x, y)\, dy\, dx$

(d) $\int_0^{\ln y} \int_1^{10} f(x, y)\, dy\, dx$

(e) $\int_0^{\ln 10} \int_0^{e^x} f(x, y)\, dy\, dx$

26. If the area of a triangle with vertices $(2, -1, -2)$, $(3, 1, -2)$ and $(3, a, 0)$ is equal to 3, then the product of all possible values of a is

(a) -15

(b) -9

(c) 15

(d) 6

(e) 9

27. $\int_0^1 \int_y^1 \cos(x^2)\, dx\, dy =$

(a) $\dfrac{\sin(1)}{2}$

(b) $\dfrac{3}{2}$

(c) $\dfrac{1}{2}$

(d) $\dfrac{\cos(1)}{2}$

(e) 0

28. $\lim_{(x,y)\to(0,0)} \frac{x^4}{x^4+y^2} =$

(a) does not exist

(b) 0

(c) 1

(d) $\frac{1}{2}$

(e) -1

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(a)**	$\lambda = \frac{15}{4}$, $(x, y) = (\frac{5}{4}, \frac{15}{4})$, $f = -\frac{75}{8}$.
2	**(c)**	$\int_1^4 2z\,dz \cdot \int_0^1 xe^{-x^2}dx = 15 \cdot \frac{1-e^{-1}}{2}$.
3	**(a)**	$\frac{dy}{dx} = \frac{t}{2} = \frac{1}{2} \Rightarrow t = 1$, $(a, b) = (4, 0)$, $a + b = 4$.
4	**(a)**	$r^2 \cos 2\theta = x^2 - y^2$, so $x^2 - y^2 - z^2 + 1 = 0$, i.e. $-x^2 + y^2 + z^2 = 1$.
5	**(d)**	$\frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}} = \frac{6}{3} = 2$, so the projection is $2\vec{v} = \langle -2, 2, 2\rangle$.
6	**(b)**	Base triangle $x + 2y \le 7$: $\int_0^7 \int_0^{(7-x)/2} \int_0^{7-x-2y} dz\,dy\,dx$.
7	**(a)**	$f_z = \sin(xy^2 + 2z) + 2z\cos(xy^2 + 2z) = -\sin 1 - 2\cos 1$ at $(1, -1, -1)$.
8	**(b)**	$\nabla F \times \nabla G \parallel (16, 13, 5)$; line $(4+16t, -3+13t, 5+5t)$, $t = -1$ gives $(-12, -16, 0)$.
9	**(d)**	Union is the sector $0 \le \theta \le \frac{\pi}{4}$, $r \le 2\sqrt{2}$: $I = \frac{\pi}{4} \cdot \frac{16\sqrt{2}}{3} = \frac{4\sqrt{2}\pi}{3}$.
10	**(d)**	The integrand y is odd and the region is symmetric about $y = 0$, so the value is 0.
11	**(c)**	$z_s = 3(x + y)e^{xy}$ with $(x, y) = (2, 0)$ gives 6.
12	**(d)**	Interior $f = -9$; boundary $f = -x^2 - 4x + 41$ has maximum 45 at $x = -2$.
13	**(b)**	$\nabla F = (3, 2, 1)$; line $(e + 3t, e + 2t, e + t)$ at $t = -1$ gives $(e - 3, e - 2, e - 1)$.
14	**(a)**	$\rho^2 = 3\rho\cos\phi \Rightarrow \rho = 3\cos\phi$.

Q	Ans	Brief explanation
15	**(b)**	Point $(0,4,0)$ gives $b=0$; point $(2,2,1)$ gives $a=1$; $a+b=1$.
16	**(b)**	$V = 2\pi \cdot \frac{8}{3} \cdot \frac{\sqrt{2}}{2} = \frac{8\sqrt{2}\pi}{3}$ with $\frac{\pi}{4} \le \phi \le \frac{\pi}{2}$.
17	**(a)**	Circle of radius 2: $A = \frac{1}{2}\int_0^{\pi} 16\cos^2\theta\, d\theta = 4\pi$.
18	**(c)**	$d^2 = x^2+y^2-6x-6y+82$, minimum at $(3,3)$: $d^2=64$, $d=8$.
19	**(d)**	Critical points $(0,1)$ ($d=-16$, saddle) and $(\frac{4}{3},\frac{7}{3})$ ($d=16$, $f_{xx}<0$, max).
20	**(c)**	$V = \int_0^{2\pi}\int_0^5 r^2\, dr\, d\theta = \frac{250\pi}{3}$.
21	**(b)**	Swap order: $\int_0^4 x\sin x\, dx = [-x\cos x+\sin x]_0^4 = -4\cos 4+\sin 4$.
22	**(c)**	Ball $\rho \le 2\cos\phi$, $0\le\phi\le\frac{\pi}{2}$; integrand $\rho^3\cdot\rho^2\sin\phi = \rho^5\sin\phi$.
23	**(c)**	$\iint e^{x+y}dA = \frac{(e-1)^2}{2}$, area $\frac{1}{2}$, average $=(e-1)^2$.
24	**(c)**	Candidates: interior value 1; edges give range $[0,9]$. So $M=9$, $m=0$, $M+m=9$.
25	**(c)**	Region $0\le x\le\ln 10$, $e^x\le y\le 10$.
26	**(a)**	$\overrightarrow{AB}\times\overrightarrow{AC} = \langle 4,-2,a-1\rangle$, $(a-1)^2=16$, $a=5,-3$, product -15.
27	**(a)**	Swap order: $\int_0^1 x\cos(x^2)\, dx = \frac{\sin 1}{2}$.
28	**(a)**	Along $y=0$ the value is 1, along $x=0$ it is 0, so the limit does not exist.